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Bijay Gaudel

Ph.D. Researcher in Robotics · Manipulation, Perception & Autonomous Navigation

Robotics researcher and engineer building perception-driven, safety-aware manipulation and navigation systems that transfer from simulation to real hardware. I develop learning-based manipulation, human-in-the-loop autonomy, and robust perception as end-to-end ROS 2 / Gazebo / deep-RL pipelines — with peer-reviewed results at IROS and ICUAS and open-source code deployed on real robots.

TECHNICAL SKILLS

PROGRAMMING	Python · C / C++ · Bash · MATLAB
ML & VISION	PyTorch · TorchRL · PyPose · TensorFlow · Keras · scikit-learn · OpenCV · Hugging Face · NumPy / SciPy
ROBOTICS & SIM	ROS 1 / ROS 2 · Gazebo (Harmonic) · Isaac Sim · MuJoCo · ManiSkill · AirSim · haptic teleoperation · reinforcement learning
INFRA & TOOLING	Docker · Apptainer · SLURM multi-GPU · Ubuntu / Linux · Git / GitHub

EXPERIENCE

2022 – Present
Hoboken, NJ

Graduate Research Assistant

Stevens Institute of Technology

- Lead Ph.D. research on deployable robotic casualty care — perception, learning-based manipulation, and human-in-the-loop autonomy for contact-rich care tasks.
- Develop image-based tactile sensing and learning algorithms for dexterous manipulation, targeting high-rate tactile perception and sim-to-real transfer in safety-critical settings.
- Build end-to-end ROS 2 / Gazebo pipelines — simulation stacks, haptic teleoperation data logging, and multi-GPU (SLURM) training — for reproducible manipulation-learning experiments.

2021 – 2022
Kathmandu, Nepal

AI Engineer

Samadhan Engineering Pvt. Ltd.

- Designed a national STEAM-aligned AI curriculum and hands-on mini-projects for grades 8–12.
- Contributed perception and control components to a search-and-rescue UAV project.

2019 – 2021
Nanjing, China

Research Assistant

Nanjing University of Aeronautics & Astronautics (NUAA)

- Researched complex-network analysis: link prediction in bipartite networks via composite similarities and node classification from local structural properties.
- Recognized with NUAA's Excellent Master's Thesis Award; results published in KSII Transactions and at the CCF Conference on Big Data.
- As Teaching Assistant (2020), designed and led a Python programming lab and mentored third-year undergraduate project teams.

SELECTED PROJECTS

MAPscoff — Safe Cross-Embodiment Grasp-Compliance Calibration
stevens-armlab/graspCompliance ↗

- Few-shot calibration of passive Cartesian grasp-compliance for unseen robot hands from only 20 wrench–displacement probes; estimates constrained symmetric positive-semidefinite to guarantee passivity for safe impedance control.
- Zero passivity violations on leave-one-hand-out evaluation (vs. 100,000+ for unconstrained baselines) while outperforming a MAML baseline; benchmark of 7 hands, 8,805 grasps, 436,609 responses.

Event-Camera Reinforcement-Learning Navigation on Jackal
gaudelbijay/event_jackal ↗

- RL navigation that learns directly from asynchronous event streams via compact event representations and contrastive perception features.
- Trained MLP / CNN / GRU / Transformer agents with Soft Actor-Critic; transferred to a real Jackal robot with minimal retuning.

Contact-Aware Human Joint-State Estimation for Casualty Manipulation
stevens-armlab/casperjointpose ↗

- Depth-only model estimating 23-DoF human joint angles and body-link classes, with feasible-state prediction returning physically valid joint sets rather than a single estimate.
- Staged training for angle prediction, ranking, and uncertainty modeling, with reproducible splits and multi-GPU (SLURM) workflows.

ROS 2 + Gazebo Harmonic Casualty-Manipulation Simulation Stack
stevens-armlab/caveman_ros2 ↗

- End-to-end ROS 2 Jazzy / Gazebo Harmonic stack simulating a Clearpath Husky with UR manipulators and a compliant human model for contact-rich research.
- Single- and dual-arm modes with a custom gripper; documented setup for cross-system reproducibility.

Diffusion-Based Defense Against Adversarial Perception Attacks
gaudelbijay/diffusion-defender ↗

- Diffusion denoising pipeline with an FFT-based runtime detector to flag and suppress online adversarial image attacks.
- Integrated with an RL attacker in ROS + AirSim by routing perception through denoised images.

Additional projects & code → gaudelbijay.github.io/projects

PUBLICATIONS

2025 — **Learning Optimal UAV Trajectory for Data Collection in 3D Reconstruction Model**
B. Gaudel, S. Jaffarnejad
Int'l Conference on Unmanned Aircraft Systems (ICUAS)

2025 — **CVD-SfM: A Cross-View Deep Front-End Structure-from-Motion System for Sparse Localization in Multi-Altitude Scenes**
Y. Li, Y. Huang, **B. Gaudel**, H. Jafarnejadsani, B. Englot
IEEE/RSJ Int'l Conference on Intelligent Robots and Systems (IROS)

2023 — **Link Prediction in Bipartite Networks Using Composite Similarities**
B. Gaudel, D. Shrestha, N. Basnet, N. Rajkarnikar, S. R. Jeong, D. Guan
KSII Transactions on Internet and Information Systems, 17(8)

2021 — **Linear Frequency-Modulated Photonic RADAR Using Injection Locking in a Semiconductor Laser**
B. Nakarmi, H. Chen, I. A. Ukaegbu, **B. Gaudel**, et al.
IEEE Int'l Conference on Electronics, Communications and Information Technology (ICECIT)

2020 **Graph Representation Learning Using Attention Networks**
B. Gaudel, D. Guan, W. Yuan, D. Shrestha, B. Chen, Y. Tu
CCF Conference on Big Data (Springer)

EDUCATION

2022 – Present **Ph.D., Mechanical Engineering — Robotics**
Stevens Institute of Technology · Hoboken, NJ, USA

2018 – 2021 **M.Eng., Computer Science — Data Science**
Nanjing University of Aeronautics & Astronautics · Nanjing, China

2013 – 2017 **B.Eng., Electrical Engineering**
Institute of Engineering, Tribhuvan University (WRC) · Pokhara, Nepal

AWARDS & HONORS

2022 **Provost Fellowship**
Stevens Institute of Technology

2021 **Excellent Master's Thesis Award**
Nanjing University of Aeronautics & Astronautics — *Local-structure-based link prediction and node classification on complex networks*